Computer Vision – 21CSE454T

Unit 3 - Session 9

SLO 1: Applications & Review of HT/GHT- Propose modernization strategies for legacy HT systems

Duration: 10 Minutes Activity

Question:

THE 6 STRATEGIES

Read all 6 strategy cards carefully before attempting Part A. Each card explains the legacy problem and how the strategy addresses it.

RND S1

Randomised GHT *[Random Sampling]*

Legacy Problem: Full accumulator voting — every edge point votes into every valid bin, causing billions of operations per frame.

How it works: Instead of using all edge points, a random subset is sampled. Only sampled points cast votes, drastically reducing total operations.

HRC S2

Hierarchical Search *[Coarse-to-Fine Refinement]*

Legacy Problem: Flat single-resolution accumulator — the system searches all bins at full resolution even when the object could be narrowed down coarsely first.

How it works: A coarse-resolution accumulator first identifies candidate regions, then a fine-resolution search is run only within those regions.

GPU S3

GPU Parallelisation *[Parallel Computing]*

Legacy Problem: Sequential CPU voting — votes are cast one edge point at a time on a single CPU thread, making real-time processing impossible for large accumulators.

How it works: All edge points cast votes simultaneously on thousands of GPU cores. Accumulator updates are parallelised across the parameter space.

HYB S4

Hybrid DL + HT Pipeline *[Deep Learning Integration]*

Legacy Problem: HT runs on raw edge maps from simple Canny detection, making it brittle in cluttered, textured, or low-contrast scenes.

How it works: A CNN pre-processes the image to produce a clean, semantically filtered feature map or edge probability map; HT then runs on this improved input.

C2F S5

Coarse-to-Fine Voting *[Adaptive Resolution]*

Legacy Problem: Uniform bin resolution — the accumulator uses the same fine resolution across all bins, wasting computation on regions that contain no object.

How it works: Voting begins with large bins to locate approximate object position, then progressively shrinks bin size only in high-vote regions.

DIM S6

Dimension Reduction *[Parameter Space Reduction]*

Legacy Problem: Full N-dimensional accumulator — all parameters (cx, cy, a, b, theta) are searched simultaneously, causing exponential memory growth.

How it works: Domain knowledge constrains or eliminates dimensions: fixing aspect ratio removes b, constraining orientation reduces theta bins, decoupling parameters enables sequential 2D searches.

PART A — QUICK RANKING TASK

Using your completed table, answer the following:

B1. Which single strategy gives the GREATEST speed improvement for a real-time iris detection system?

Strategy: GPU Parallelisation Reason: Thousands of GPU cores cast votes simultaneously, enabling real-time voting.

It removes the sequential CPU bottleneck while preserving the full voting process.

B2. Which strategy has the SMALLEST trade-off (i.e. least downside)? Justify your choice.

Strategy: Hierarchical Search Reason: It narrows the search coarsely first, then refines only candidate regions.

This reduces computation without discarding edge points or permanently removing parameter dimensions.

B3. Which TWO strategies work best when combined together? Name the combination and explain the synergy.

Strategies: GPU Parallelisation + Dimension Reduction

GPU parallelisation speeds up voting, while dimension reduction shrinks the parameter space.

Together they reduce both computation time and memory demand.

B4. A legacy iris detection system runs at 2 fps on CPU and consumes 800 MB RAM. Which strategies would you apply FIRST and SECOND to fix this? Explain.

First: GPU Parallelisation Second: Dimension Reduction

GPU addresses the 2 fps CPU bottleneck; dimension reduction addresses the 800 MB memory cost.

PART B — Reflection

PART B — REFLECTION

R1. All 6 strategies introduce a trade-off. Which trade-off do you consider MOST acceptable in a safety-critical medical imaging system? Why?

A coarse-to-fine search is an acceptable trade-off because it reduces computation while retaining fine-resolution refinement in candidate regions.

It does not discard all detailed evidence and can be checked at the fine stage.

R2. A colleague argues: "Just replace legacy HT with a CNN — no modernisation needed." Write ONE strong counter-argument defending the value of modernised HT.

Modernised HT can remain interpretable and geometry-based while becoming faster and more robust; it can also complement CNN features rather than requiring a full replacement.

R3. Dimension Reduction requires domain knowledge to constrain parameters. Name ONE scenario where this knowledge is unavailable and explain what strategy you would use instead.

If object shape, aspect ratio, and orientation are unknown, use Hierarchical Search or GPU Parallelisation instead of fixing parameters; these do not require removing dimensions.